

What Do Relation Labels Add to Image–Text Adaptation?

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Abstract

An image search system should reject wrong descriptions while recognizing different correct descriptions. Training labels can mark a caption as supported, contradicted, or unresolved by an image. We test what these labels add beyond extra correct examples. Twelve training rules use one pretrained model’s fixed features, with matched caption pools and tuning budgets. Adding supported captions improves the annotated statement test by 5.81 percentage points. It gains only 0.14 points on edited captions and loses 8.56 points when retrieving captions for images. Gains concentrate on rejecting added objects and attributes; combined replacement and swap edits decline. Combining stronger contradiction penalties with removal of unresolved pairs provides no clear further gain. Similarity weights do not consistently beat simple controls. Two exact identities explain how these rules change the training targets and penalties on other captions. The comparison tests which benefits carry beyond the annotated task.

1 Introduction

A person searching a photo collection may describe the same scene in several ways. A photo of a child holding a red ball can match both “a child with a red ball” and “a red ball held by a child.” It should not match “a child with a blue ball.” A useful image–text model must recognize valid descriptions while distinguishing small changes that make them incorrect.

CLIP learns image and text vectors by scoring associated image–caption pairs above other candidates (Radford et al., 2021). Those other captions are not necessarily wrong. Some describe the image correctly, while others mention details the image cannot establish. This motivates three relation labels: *supported*, *contradicted*, and *neutral*. Here, neutral means that the image leaves the statement unresolved. Such labels suggest a natural training

policy: add supported captions to the positive targets, increase the penalty on contradictions, and omit neutral pairs.

Each operation also changes the learning problem. Adding positives changes the target distribution. Weighting negatives changes their relative and total gradient contributions. Omitting pairs removes competition from the loss. An improvement over ordinary training therefore need not come from the specific semantic distinction that motivates a policy. The relevant question is: *what do relation labels add beyond extra positive targets, under the same data, model, and selection budget?*

Prior work gives the necessary comparisons. Kamath et al. (2024) show that gains against difficult wrong captions can hide sensitivity to valid rewordings, including after generic positive augmentation. FSC-CLIP studies preservation of broader multimodal capabilities (Oh et al., 2024), and CLIC evaluates efficient adaptation on both discrimination and retrieval (Peleg et al., 2025). These findings make retention a necessary outcome of a relation-supervision study.

We reuse one annotated caption pool to compare positive expansion, contradiction weighting, neutral exclusion, and their controls. All policies train the same small adapters over frozen CLIP features. This design connects an intervention-level comparison to three practical tests: rejecting wrong captions, accepting alternative valid captions, and retrieving a match from a large candidate pool. The contribution is a controlled account of what the relation policy adds, supported by per-seed results and an analytical description of how its targets change the update.

2 A Controlled Study of Relation Labels

One candidate pool, three interventions. Every training image has its source captions and statements with relation labels from e-SNLI-VE

(Kayser et al., 2021). All contrastive policies use the same candidate pool. *Source positives* treats the five original captions as positive targets and all statement candidates, including supported ones, as negatives. *Positive expansion* treats every label-supported statement as a positive. Starting from that expanded set, *contradiction weighting* doubles the weight of labeled contradictions in the loss, and *neutral exclusion* removes labeled neutral pairs. *Both* combines those changes. Unannotated cross-image pairs stay eligible negatives. Neutrality is a relation label, not a model confidence estimate.

We include count-matched random exclusion, target smoothing, and supervised pairwise ranking. Four additional policies test text-similarity weighting: a fixed threshold, its shuffled-weight control, a batch-median threshold, and a constant negative weight. The shuffle preserves the batch-wide weight distribution while breaking its association with individual pairs. There are twelve trainable configurations. Appendix A gives their exact rules and archive names.

Common objective. Let s_{ij} be the scaled cosine score of image i and caption j . For image i , let P_i be its positive-caption set and w_{ij} a nonnegative candidate weight. Positive weights are one; zero omits a pair. The image-to-text loss is

$$\ell_i = \log \left(\sum_j w_{ij} e^{s_{ij}} \right) - \frac{1}{|P_i|} \sum_{j \in P_i} s_{ij}. \quad (1)$$

It gives equal target probability to each positive, following the mean-log-probability form of multi-positive learning (Khosla et al., 2020). We average it with the corresponding text-to-image loss over captions having a positive image. The ranking control uses supported–contradicted pairs instead. Appendix A defines rectangular batches and the smoothing target.

Training and selection. We use frozen OpenCLIP ViT-B/32 features from the LAION-2B checkpoint. Two zero-initialized residual maps add learned linear corrections to the features, with 524,288 trainable parameters; the encoders and score scale remain fixed. The image-disjoint split contains 1,200 training, 100 calibration, 100 validation, and 400 test images.

Each configuration runs for ten epochs at three learning rates and three seeds, giving 108 runs. A fixed validation score averages retrieval accuracy at

rank one and supported-versus-contradicted statement accuracy. Validation retrieval counts source or supported captions as correct. Selection includes epoch zero. One learning rate per policy is chosen across all three seeds, and each seed retains its best validation epoch. We then evaluate the resulting 36 checkpoints and the frozen reference. Appendix C specifies the recorded protocol.

Four complementary measurements. On e-SNLI-VE, we measure the fraction of supported statements ranked above contradicted statements, averaged by image. SugarCrepe tests whether an original correct caption outranks an edited wrong caption, or *foil*, over all 7,511 official pairs (Hsieh et al., 2023). SugarCrepe++ adds a second valid caption and requires both positives to beat the foil over 4,757 triplets (Dumpala et al., 2024). COCO retrieval uses 5,000 images and 25,000 captions. Recall at rank one (R@1) is the percentage of queries whose highest-scoring candidate is an annotated match. We evaluate image-to-text (I→T) and text-to-image (T→I) retrieval on the full pool. The two discrimination measures are primary; retention measures are secondary. Primary intervals use 10,000 paired resamples of images with a Bonferroni adjustment for six planned contrasts.

3 What the Controls Reveal

Positive expansion mainly improves the annotated task. Relative to source positives, adding supported captions gains 5.81 percentage points on e-SNLI-VE, but only 0.14 on SugarCrepe. Table 1 separates the two negative-pair interventions while keeping the expanded positive set fixed. The combined policy differs from positive expansion by -0.26 points on e-SNLI-VE and -0.16 on SugarCrepe. Their adjusted 99.17% image-bootstrap intervals are $[-0.60, 0.07]$ and $[-0.51, 0.21]$, respectively. Neither comparison establishes an additional gain from combining the relation-specific rules.

The aggregate transfer score hides opposite effects. SugarCrepe edits either add content, replace it, or swap its assignment. Table 2 shows that positive expansion helps reject captions with added objects or attributes. Its mean falls on the combined replacement/swap group. Both directions hold in every seed. The complete seven-category breakdown is in Appendix E.

Policy	e-SNLI-VE	SugarCrepe	SC++ both	COCO I→T	COCO T→I
Frozen	79.47	81.21	68.95	56.32	39.35
Source positives	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40	55.85	39.38
Positive expansion	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23	47.29	34.84
+ contradiction weighting	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30	47.25	34.81
+ neutral exclusion	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66	48.01	35.39
+ both	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61	47.26	35.27

Table 1: Relation interventions with the same candidate pool and tuning budget. Scores are percentages; subscripts show sample standard deviation across three selected seeds for the primary endpoints. COCO columns report full-pool R@1. The final three rows add rules to positive expansion.



Source positives: A young girl is preparing to blow out her candle.

Positive expansion: a woman is making some food in a bowl

Figure 1: Top retrieved captions for the same image query (seed 17). The best annotated caption moves from rank 1 to 6. Both models search the full caption pool.

Edit group	n	Source	Expand.	Δ
Added content	2,754	88.09	90.01	+1.92
Replace/swap	4,757	80.46	79.56	-0.90
All edits	7,511	83.26	83.39	+0.14

Table 2: SugarCrepe accuracy by edit group. Differences are positive expansion minus source positives, in percentage points. Groups partition the official pairs.

A small transfer gain has a large retrieval cost. Compared with source positives, positive expansion loses 8.56 points of COCO image-to-text R@1. That is 428 fewer image queries returning a correct caption first per 5,000 queries, averaged across seeds. Text-to-image R@1 falls by 4.54 points as well. Figure 1 gives a concrete image-query failure. These full-pool tests connect the target change to the two directions of a search task.

Valid descriptions reveal the retention problem. Figure 2 compares two correct captions against the

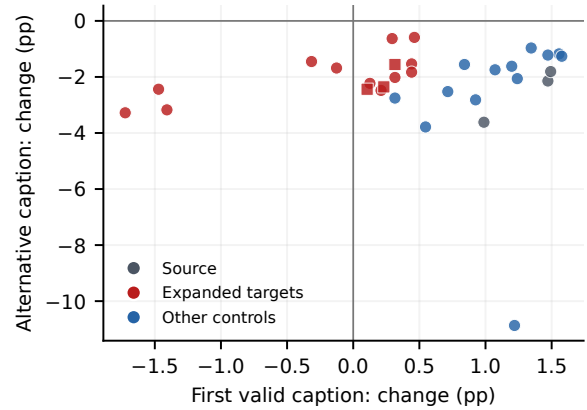


Figure 2: Changes on the same SugarCrepe++ triplets relative to the frozen model. Each point is one selected nonzero-epoch adapter. All 31 lose accuracy on the alternative valid caption; 26 improve on the first. The 5 epoch-zero selections remain at the origin and are omitted.

same foil. The effect extends across target and weighting policies, including source-positive adaptation. A model can become better at rejecting a foil using the first wording while becoming worse with a second valid wording. Requiring both positives to beat the foil therefore tests a capability missed by first-positive accuracy alone.

Neutral exclusion improves on random removal. Neutral exclusion outperforms count-matched random exclusion by 0.40 points on e-SNLI-VE and 1.77 on SugarCrepe. It nevertheless has a lower e-SNLI-VE mean than positive expansion and the same rounded SugarCrepe mean. Its advantage over random removal does not translate into a stronger overall relation policy. The remaining controls and per-seed selection outcomes appear in Appendix E.

Holding contradiction weighting in place, the selected policy with neutral exclusion scores -0.53

points lower on e-SNLI-VE than the policy that retains neutral pairs. The adjusted interval is $[-0.89, -0.19]$. This is the only contrast in the six-comparison primary family whose interval excludes zero. Thus neutral exclusion beats random removal but adds no observed advantage to contradiction weighting on the supervised distinction.

Pairwise ranking retains epoch zero for all three seeds, so its selected outcomes equal the frozen reference. Unlike the contrastive policies, this control trains only on within-image supported-contradicted pairs. Appendix E reports the full validation histories.

4 What the Objective Changes

Positive labels redistribute a fixed target budget. Equation 1 distributes one unit of target probability across the positive set. Suppose K captions are positive and M already-present candidates become additional positives. Hold the scores, candidate set, and unit weights fixed. For an original positive p , the loss derivative changes by

$$\frac{\partial \ell_{\text{expanded}}}{\partial s_{ip}} - \frac{\partial \ell_{\text{source}}}{\partial s_{ip}} = \frac{M}{K(K+M)}. \quad (2)$$

At the score level, expansion reduces each original positive’s target pressure by this amount. The new positives gain target mass at the originals’ expense, even when every added label is correct. For example, five original captions and five added positives halve each original target probability from 0.2 to 0.1 without changing any score. Appendix H derives the identity and contrasts summed-positive probability. This motivates measuring retention alongside discrimination; the learned representation also depends on shared parameters and checkpoint selection.

Negative weights mix strength and allocation. For weights held fixed when differentiating, let the nonempty set N_i contain image i ’s nonpositive candidates. At fixed scores and positive set, define

$$c_i^* = \frac{\sum_{j \in N_i} w_{ij} e^{s_{ij}}}{\sum_{j \in N_i} e^{s_{ij}}}. \quad (3)$$

Replacing every negative weight by c_i^* leaves the denominator in Equation 1 unchanged. It preserves the row loss and, holding c_i^* fixed during differentiation, every positive gradient and the total negative gradient. Only the allocation of negative gradients changes. A lower weighted loss can therefore

Weights	e-SNLI-VE	Sugar	I→T
Fixed similarity	79.56	81.80	54.33
Shuffled weights	83.34	83.49	52.27
Median similarity	82.66	83.73	53.97
Constant weights	82.15	83.67	55.21

Table 3: Similarity weights and controls, using source positives. Entries are mean percentages; the final column is COCO R@1.

reflect weaker overall competition without showing useful pair-specific assignments. Appendix H proves the identity and shows why reducing total negative gradient need not reduce every negative group.

This identity defines an analytical reference. The empirical constant control instead fixes every negative weight at 0.25. The shuffled control retains the batch-wide distribution of similarity weights. These controls ask whether identifying which pair gets which weight improves performance beyond nonspecific suppression. Table 3 shows that semantic assignment does not consistently beat its controls. Fixed-threshold weights trail their shuffle on both primary tasks. Median weighting and constant suppression differ by only 0.06 points on SugarCrepe. The constant rule retains substantially more retrieval accuracy than positive expansion. Pair-specific weighting must therefore be assessed separately from the general effect of weakening negative competition.

5 Implications for Image Search

A useful search evaluation should test multiple valid wordings and preserve the full retrieval pool. The validation score here already includes retrieval, but on a small pool from the annotation source. That does not ensure retention on a larger pool from another source. An application-oriented extension can select checkpoints on representative images and queries, then compare discrimination at a fixed acceptable retrieval loss. Reporting the two retrieval directions separately keeps a gain in one from hiding a loss in the other.

A next comparison can retain frozen-model retrieval and use relation-trained adapters only to reorder its candidate set.

Limitations

This study concerns one frozen OpenCLIP checkpoint and linear residual adaptation on a subset of e-SNLI-VE. It does not establish the behavior of full-encoder training, other data scales, or other model families. The relation labels partly inherit text-inference judgments; the release does not provide independent visual verification for every pair. Caption ownership also supplies incomplete retrieval relevance.

The multi-positive objective averages positive log probabilities. The summed-positive alternative in Appendix H is an analytical comparison. Independent policy selection changes the learning rate and retained epoch, so the comparisons estimate selected training procedures. Image-bootstrap intervals condition on three seeds and do not quantify uncertainty over new training runs. Sugar-Crepe++, retrieval, weighting controls, and edit-group summaries are secondary or descriptive. COCO-derived benchmarks overlap, so they measure different capabilities rather than independent replications. Appendix B records source and execution details.

Ethical Considerations and Assistance

We use existing public datasets under their source terms and collect no new human judgments. Captions and pretrained features may reflect source biases. Generative AI assisted planning, implementation, analysis, literature discovery, and drafting. The authors are responsible for checking the released computations and manuscript.

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A Complete Policy Definitions

A batch has B unique image rows and T text columns, containing all selected source and hypothesis captions for its images. The relation matrix $R \in \{0, 1, 2, 3, 4\}^{B \times T}$ uses codes for unannotated, source, supported, contradicted, and neutral pairs, respectively. Relations apply only to their annotated image–text pair. An absent cross-image annotation is an eligible negative, not a neutral label.

Let P_{ij} be the binary positive mask. Every contrastive policy includes source captions. Expanded policies additionally include supported hypotheses. Candidate weights start at one and are changed according to Table 4; all positive weights are then forced to one. The scalar loss averages eligible image rows and eligible text columns independently, with weight $1/2$ on each direction. The reverse direction transposes the same mask and weight matrices. Columns without a positive image are omitted from the reverse average. These rules preserve the meaning of rectangular batches with several captions per image.

Archive name	Positive set	Weight or target rule
clip	Source	All weights one
multipositive	Source + supported	All weights one
grounded	Source + supported	Contradictions: 2; neutrals: 0
grounded_no_hardenning	Source + supported	Neutrals: 0
grounded_no_abstention	Source + supported	Contradictions: 2
random_exclusion	Source + supported	Count-matched random exclusions
smoothing	Source	$0.9 \text{ Unif}(P) + 0.1 \text{ Unif}(J)$
constant	Source	All negatives: 0.25
sanw_fixed	Source	Similarity kernel, threshold 0.5
sanw_median	Source	Similarity kernel, batch median threshold
shuffled	Source	Permuted fixed-threshold negative weights
pairwise_rank	Supported/contradicted	Image-balanced pairwise softplus

Table 4: All twelve trainable policies. A weight of zero exactly excludes a pair. The frozen reference has no trained parameters.

Detached similarity weights. Let t_j be the unit frozen text feature for candidate j , and C_i the source-caption indices of image i . Define

$$a_{ij} = \frac{1 + |C_i|^{-1} \sum_{k \in C_i} t_k^\top t_j}{2}. \quad (4)$$

The average of source features is not renormalized. The negative weight is

$$w_{ij} = \text{sigmoid}\left(\frac{\tau - a_{ij}}{0.05}\right). \quad (5)$$

The fixed rule sets $\tau = 0.5$. The median rule uses the statistical median of a_{ij} over nonpositive cells of the current batch, averaging the two central values for an even count. Positive weights remain one. All features, thresholds, and weights in this calculation are detached from the training graph. The kernel never uses adapted features.

Shuffling and random exclusion. The shuffled rule permutes the complete batch-wide multiset of negative weights from the fixed-threshold rule, leaving positive weights intact. It preserves the weight distribution but need not preserve each row’s weighted mass. Random exclusion samples nonpositive cells without replacement within each image row, matching that image’s number of neutral annotations. Because the reverse direction uses a transpose, its per-text exclusion counts need not match the neutral rule. Both randomizations use a seeded generator whose state advances through batches and epochs.

Smoothing and ranking. Smoothing uses source-positive targets and unit weights. Its row

target is $0.9 \text{ Unif}(P) + 0.1 \text{ Unif}(J)$, where J contains every candidate. For ranking, let E_i and C'_i denote supported and contradicted hypotheses of image i . The eligible-image loss is

$$\ell_i^{\text{rank}} = \frac{\sum_{e \in E_i} \sum_{c \in C'_i} \text{softplus}(s_{ic} - s_{ie})}{|E_i| |C'_i|}. \quad (6)$$

We average over images with both sets nonempty. Ranking has no reverse direction and does not train on source, neutral, or unannotated pairs.

B Data, Provenance, and Reconstruction

The current execution was reconstructed from retained methods and proof text after the earlier workspace became unavailable. Historical aggregate tables are kept in a separate context directory and are not inputs to training, evaluation, or paper tables. The exact old similarity kernel and the base target set for smoothing were not recoverable. The explicit definitions in Appendix A were fixed before the new runs and are documented in the reconstruction record. The present study is a new execution under those recorded choices.

B.1 Relation annotations and source captions

We use the official e-ViL release of e-SNLI-VE at revision 94f17f5c1895. Its training, development, and test sources contain 401,717, 14,339, and 14,740 annotated hypotheses over 29,783, 1,000, and 1,000 images, respectively. The official pools have no shared image IDs or annotation-pair IDs. With Python’s `random.Random(42)`, we sample from sorted image IDs: 1,200 training images,

Split	Images	Support	Contradict	Neutral
Train	1,200	5,319	5,846	5,027
Calibration	100	522	554	355
Validation	100	502	534	339
Test	400	2,088	2,278	1,495

Table 5: Retained hypothesis annotations. Each image also has five source captions. All validation and test images, and 1,192 of 1,200 training images, have both supported and contradicted hypotheses.

200 development images split into 100 calibration and 100 validation images, and 400 test images. All annotations of each selected image remain in source order. Table 5 gives the resulting counts.

The source captions come from the official Flickr30k Entities annotation archive at revision 68b3d6f12d1d. We use its five sentence-file lines per image in stored order. Entity markup is removed while retaining source words, case, and punctuation, with single spaces between stored tokens. This source covers all 31,783 images. An incomplete alternative caption mirror covers only 31,014 and was not used. The image bytes come from the `nlphuji/flickr30k` mirror at revision 2b239befc81b. Each selected image passes ZIP size/CRC checks, JPEG verification, and full decoding. No selected image is silently skipped.

An earlier deprecated relation release was rejected during an input-only audit because of development/test overlap. This decision preceded feature-based evaluation. The chosen official release, exact IDs, source URLs, full revisions, file hashes, and selection order are recorded in the per-dataset provenance files. The released labels include inherited text-inference judgments, so their names are operational annotation categories rather than a claim of new visual adjudication.

B.2 Transfer benchmarks

SugarCrepe uses all 7,511 pairs over 1,560 images from revision 0047054b2439. Its category counts are 692 added attributes, 2,062 added objects, 788 replaced attributes, 1,652 replaced objects, 1,406 replaced relations, 666 swapped attributes, and 245 swapped objects. SugarCrepe++ uses all 4,757 triplets over 1,542 images from revision c083bbf039a5; it covers the five replacement/swap categories with the same category counts.

Each benchmark preserves its own released text strings. SugarCrepe++ changes 1,203 first-positive strings and 878 foil strings relative to the matched SugarCrepe records. Its first-positive accuracy therefore cannot substitute for SugarCrepe’s replacement/swap score. Two released SugarCrepe++ triplets have identical first-positive and foil strings, making a strict win impossible; these remain in the official denominator. One triplet repeats the same string in both positive positions. No item is edited or removed in response to model scores.

COCO uses all 5,000 Stanford Karpathy test images. The source contains 25,010 captions; taking the first five in stored order for every image yields 25,000 candidates and omits ten sixth captions. Caption strings are not normalized. The COCO JPEG cache contains 6,377 distinct benchmark images. It uses the official `val2014` source for Karpathy members and `val2017` for remaining benchmark images. All 183 shared SugarCrepe/Karpathy images were checked across these source versions and are byte-identical.

B.3 Overlap and duplicate audit

All 1,542 SugarCrepe++ images also occur in SugarCrepe. COCO retrieval shares 183 images with SugarCrepe and 182 with SugarCrepe++. These are complementary measurements on overlapping pools. The selected Flickr30k images share neither IDs nor image-byte hashes with the COCO-derived benchmarks. The four e-SNLI-VE splits also have no shared image bytes.

The input audit finds 48 repeated exact image/text groups in e-SNLI-VE, each appearing twice. Of these, 46 are repeated hypothesis groups. Their relation labels do not conflict. All rows are retained, so repeated annotations retain their multiplicity in the loss and metrics. The two positive/foil equality cases account for the within-triplet relation conflicts in SugarCrepe++. The audit records source defects before scoring rather than changing benchmark contents.

B.4 Frozen feature extraction

The encoder is `laion/CLIP-ViT-B-32-laion2B-s34B-b79K`, revision `1a25a446712b`, with weight SHA256 beginning `ac4f8c4b88af`. We use OpenCLIP 2.32.0 and PyTorch 2.7.1 on CPU. Images are resized with bicubic interpolation, center-cropped to 224×224 , converted to RGB, and normalized with the checkpoint’s channel means and standard devia-

tions. The 512-dimensional image and text outputs are stored in float32 after unit L2 normalization. The fixed scale is 100.00000762939453.

Text encoding trims only causal padding positions after the last end-of-text token in a batch. A parity check compares normalized outputs against the stock encoder with tolerance $2 \cdot 10^{-6}$. Recorded per-dataset parity checks, preprocessing parameters, source hashes, input-manifest hashes, feature-cache hashes, and full model pins are stored in each feature metadata file. Reused features require exact image-byte or text-string identity under the same encoder and preprocessing.

C Training and Checkpoint Selection

Let $x, z \in \mathbb{R}^{512}$ be unit frozen image and text features. The trainable maps are

$$\hat{x} = \frac{x + A_I x}{\|x + A_I x\|_2}, \quad \hat{z} = \frac{z + A_T z}{\|z + A_T z\|_2}. \quad (7)$$

Both 512×512 matrices start at zero and have no bias. Their total parameter count is $2 \cdot 512^2 = 524,288$. The score is $s_{ij} = \alpha \hat{x}_i^\top \hat{z}_j$, where $\alpha > 0$ is the pretrained checkpoint’s fixed scale. Feature extraction and source hashes identify the exact encoder and input order.

The learning-rate grid is $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ and training seeds are 17, 29, 43. Each candidate trains for ten epochs with image batches of 32. All source captions and hypotheses assigned to those images enter the text candidate pool. AdamW uses weight decay 0.01, betas (0.9, 0.999), and epsilon 10^{-8} . The gradient norm is clipped at one. There is no warmup or learning-rate schedule. Separate random generators control image order and stochastic policy operations, so weight shuffling does not alter batch order.

Validation retrieval uses every validation caption and hypothesis as a candidate. Both source captions and supported hypotheses are relevant matches. Its R@1 and the image-balanced supported-versus-contradicted accuracy receive equal weight in the selection score. The epoch with the highest score is retained for each seed and learning rate, including epoch zero. Exact ties retain the earliest epoch. A policy selects one learning rate by averaging these best scores over all three seeds; exact ties select the smaller rate. Held-out scoring starts after all 108 candidates and 36 selections are frozen.

Ranking evaluation accumulates dot products in float64 over adapted float32 normalized features.

Exact retrieval-score ties follow candidate manifest order; relation-ranking ties receive half credit. The fixed positive score scale is omitted from ranking metrics because it cannot change the order. Images without both supported and contradicted hypotheses are omitted only from the relation average, with their count recorded.

An immutable execution ledger binds the protocol, code, manifest, feature metadata, cache, and split IDs before fitting. Every candidate saves its best state, all eleven validation evaluations, and a completion record with artifact hashes. The root epoch-zero state is retained separately. Resume accepts a completed candidate only when its identities and hashes match. Interrupted candidates restart deterministically from epoch zero. Selected checkpoints are copied into a separate tree while all candidate states remain available.

D Metrics and Statistical Comparisons

For e-SNLI-VE, compare every supported hypothesis against every contradicted hypothesis within an image. A larger supported score receives one point and an exact tie receives half a point. Average pairs within each eligible image, then average images. This gives each image equal weight despite different numbers of annotated statements.

SugarCrepe uses a strict positive-score comparison against its foil. SugarCrepe++ reports each positive separately and their conjunction against the same foil. The conjunction is one only if both scores are strictly greater. Per-category results retain official labels. The addition group pools added objects and attributes; the combined replacement/swap group pools the remaining categories. Grouped scores use item counts as weights, not an unweighted mean of category percentages.

COCO image-to-text R@1 counts an image query as correct when its highest-scoring caption belongs to the image’s annotated caption set. Text-to-image R@1 requires the source image at rank one. All candidates remain in the full pool during scoring. Source-caption ownership is the operational relevance label.

The six primary contrasts compare the combined relation policy against positive expansion, neutral exclusion alone, and contradiction weighting alone, on e-SNLI-VE and SugarCrepe. For each contrast, average the paired outcomes across the three selected seeds before sampling images. Each bootstrap draw samples images with replacement and

retains all associated items. With 10,000 replicates, seed 20261003, and family size six, two-sided percentile bounds use tail probability $0.05/(2 \cdot 6)$, yielding 99.1667% intervals. They condition on the selected runs and data images. Secondary metrics and category groupings are descriptive.

E Full Fresh Results and Execution Checks

Tables 6–13 give all selected-policy means, seed outcomes, primary intervals, category scores, and retrieval transitions. Five selected states retain epoch zero: all three ranking runs and fixed-similarity seeds 29 and 43. Those states have predictions identical to the frozen reference.

The illustrative query in Figure 1 is coco 60623. It is the first image in COCO manifest order that source-positive adaptation retrieves correctly at rank one in all three seeds, while positive expansion fails at rank one in every seed and falls outside the top five in seed 17. The displayed captions are the two seed-17 top results. This deterministic post hoc choice illustrates a recorded failure and is not a representative sample or an extra significance test.

F Separate Relation-Estimator Diagnostic

This diagnostic predicts released supported, contradicted, and neutral labels from frozen features. It does not provide labels or weights to the adapter study. Source captions and unannotated cross-image pairs are excluded. Each annotated image/hypothesis pair is one example, and each image remains entirely within its training, validation, calibration, or test split.

Features and fitting. The image-only and text-only models use the corresponding 512-dimensional features. The multimodal model concatenates image, text, their elementwise product, and their absolute difference. A standard scaler is fitted only on training rows. Multinomial logistic regression uses an intercept, L2 regularization, `lbfgs`, tolerance 10^{-6} , no class weights, and seed 17. The grid $C \in \{0.01, 0.1, 1, 10\}$ is selected by validation macro-F1, with smaller C breaking ties. There is no refit on validation data.

The original reconstruction set a 2,000-iteration cap. Text fitting reached that cap, so a separately recorded amendment raised the cap to 10,000 without changing the model, grid, splits, or tolerance. The amended run has its own ledger and output

directory; the original diagnostic remains archived. This numerical-completion amendment does not alter the frozen primary-adapter protocol.

Calibration and selection gates. Sort calibration image IDs and permute them with NumPy seed zero. The first 50 images fit a positive scalar temperature by minimizing mean three-class negative log likelihood; the other 50 select gates. Temperature lies in $[0.05, 20]$, optimized in log space, with one and both endpoints also considered explicitly. It preserves the classifier’s argmax predictions.

For supported and contradicted predictions separately, a gate requires that class to be the argmax and its calibrated probability to meet or exceed a threshold. Search complete confidence-tie groups and choose maximum coverage among thresholds accepting at least 20 calibration pairs with a two-sided 95% Wilson lower endpoint of at least 0.90. If no threshold qualifies, the gate accepts nothing. This calibration criterion is not a simultaneous guarantee after threshold search. Final precision and coverage are measured on held-out test images. Coverage divides accepted count by all test hypotheses; precision divides correct accepted predictions by accepted count. An empty gate has undefined precision.

Metrics and out-of-fold predictions. The diagnostic reports accuracy, three-class macro-F1, 10-bin expected calibration error, Brier score, and negative log likelihood. Brier score sums squared error over the three classes before averaging examples. Calibration bins have equal width and include probability one in the last bin. Class ties follow the fixed order supported, contradicted, neutral. Nonempty gate precision receives a descriptive 95% image-bootstrap interval from 10,000 replicates with seed 20261003. Empty-acceptance replicates are counted and omitted from the percentile calculation.

For reusable training predictions, sort training image IDs, permute with seed 17, and divide them into three image-disjoint folds. Each fold trains on the other two folds, selects C on the external validation split, and calibrates on the external calibration halves. Thus held-out training labels do not enter the model or selection decisions producing their predictions. These predictions are archived for future work and do not enter adapter fitting.

Policy	e-SNLI-VE	SugarCrepe	SC++ both
Frozen	79.47	81.21	68.95
Source	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40 $\pm$ 0.84
Expanded	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23 $\pm$ 0.34
Contradiction	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30 $\pm$ 0.28
Neutral	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66 $\pm$ 0.52
Combined	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61 $\pm$ 0.45
Random	86.26 $\pm$ 0.28	81.62 $\pm$ 0.70	65.93 $\pm$ 0.57
Smoothing	79.23 $\pm$ 4.85	82.28 $\pm$ 2.18	65.97 $\pm$ 4.59
Ranking	79.47 $\pm$ 0.00	81.21 $\pm$ 0.00	68.95 $\pm$ 0.00
Fixed	79.56 $\pm$ 0.15	81.80 $\pm$ 1.01	68.00 $\pm$ 1.65
Shuffled	83.34 $\pm$ 0.57	83.49 $\pm$ 0.36	66.90 $\pm$ 0.26
Median	82.66 $\pm$ 0.19	83.73 $\pm$ 0.17	68.01 $\pm$ 0.07
Constant	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37 $\pm$ 0.48

Table 6: All selected policies on the three discrimination endpoints. Entries are percentages with sample seed standard deviations. Source, Expanded, Contradiction, Neutral, and Combined correspond to the first five trainable policies in Table 1. Other names follow Table 4.

G Reproducibility and Extensions

The archive separates source acquisition, normalized feature extraction, fit/selection, held-out prediction, statistical analysis, and manuscript generation. Manifests give stable IDs and input hashes. Candidate checkpoints and validation histories establish which states were eligible for selection. Selected predictions support metric reconstruction without rerunning training. Result tables are generated from the fresh analysis artifact rather than entered from the historical manuscript.

Extensions retained for future work include a second backbone, full-encoder training, more training data and seeds, independently image-adjudicated relations, shared-final-epoch contrasts, denominator-matched training controls, hardness-matched exclusion, the summed-positive objective, predicted relations, inverse-multiplicity weighting, retention-constrained selection, feature distillation, and many-to-many retrieval relevance. These are distinct from the completed comparisons reported in the fresh results.

H Analytical Identities and Their Scope

This appendix states the complete assumptions for the two main-text identities. They concern derivatives with respect to fixed scores. The proofs do not depend on experimental outcomes.

H.1 Notation and the common gradient

Let J be a finite candidate set, $P \subseteq J$ a nonempty positive set, and $N = J \setminus P$. Each score $s_j \in \mathbb{R}$ is finite. Weights $w_j \geq 0$ are fixed for differentiation and satisfy $w_p = 1$ for $p \in P$. Define

$$D = \sum_{j \in J} w_j e^{s_j}, \quad q_j = \frac{w_j e^{s_j}}{D}, \quad (8)$$

$$y_j = \frac{\mathbf{1}\{j \in P\}}{|P|}, \quad \ell = \log D - \sum_j y_j s_j. \quad (9)$$

Since P is nonempty and scores are finite, $D > 0$. A zero weight removes a candidate from the denominator, while its target is also zero.

Lemma 1 (Fixed-weight row gradient). *For each $j \in J$, $\partial \ell / \partial s_j = q_j - y_j$.*

Proof. Differentiating the finite sum gives $\partial D / \partial s_j = w_j e^{s_j}$. The logarithm contributes $w_j e^{s_j} / D = q_j$, and the linear target term contributes $-y_j$. $\square$

An individual positive derivative need not be negative. If $P = \{1, 2\}$ and $q = (0.8, 0.1, 0.1)$, the gradient is $(0.3, -0.4, 0.1)$. However,

$$\sum_{p \in P} \frac{\partial \ell}{\partial s_p} = - \sum_{n \in N} q_n \leq 0. \quad (10)$$

Thus claims about total positive attraction do not imply a sign for every positive score.

Policy	SC++ first	SC++ alternative	COCO I→T	COCO T→I
Frozen	79.46	75.01	56.32	39.35
Source	80.78 $\pm$ 0.29	72.48 $\pm$ 0.96	55.85 $\pm$ 0.53	39.38 $\pm$ 0.32
Expanded	79.68 $\pm$ 0.11	72.89 $\pm$ 0.49	47.29 $\pm$ 1.78	34.84 $\pm$ 0.89
Contradiction	79.72 $\pm$ 0.16	72.92 $\pm$ 0.49	47.25 $\pm$ 1.94	34.81 $\pm$ 0.92
Neutral	79.72 $\pm$ 0.33	73.64 $\pm$ 0.68	48.01 $\pm$ 0.15	35.39 $\pm$ 0.23
Combined	79.56 $\pm$ 0.36	73.64 $\pm$ 0.70	47.26 $\pm$ 1.17	35.27 $\pm$ 0.44
Random	77.93 $\pm$ 0.17	72.04 $\pm$ 0.46	43.43 $\pm$ 1.23	29.95 $\pm$ 0.65
Smoothing	80.88 $\pm$ 0.18	70.56 $\pm$ 5.56	53.85 $\pm$ 2.54	37.08 $\pm$ 3.15
Ranking	79.46 $\pm$ 0.00	75.01 $\pm$ 0.00	56.32 $\pm$ 0.00	39.35 $\pm$ 0.00
Fixed	79.64 $\pm$ 0.32	73.74 $\pm$ 2.18	54.33 $\pm$ 3.45	36.86 $\pm$ 4.31
Shuffled	80.11 $\pm$ 0.31	72.31 $\pm$ 0.15	52.27 $\pm$ 0.29	38.19 $\pm$ 0.12
Median	80.50 $\pm$ 0.18	73.37 $\pm$ 0.10	53.97 $\pm$ 0.40	39.21 $\pm$ 0.27
Constant	80.84 $\pm$ 0.16	73.60 $\pm$ 0.58	55.21 $\pm$ 0.14	39.39 $\pm$ 0.09

Table 7: Valid-caption and retrieval retention. Entries are percentages with sample seed standard deviations.

H.2 Expansion of the positive set

Proposition 1 (Target redistribution). *Assume unit weights on every candidate and fixed scores. Let P contain $K \geq 1$ positives. Relabel a disjoint set $A \subseteq J \setminus P$ of $M \geq 0$ already-present candidates as positive, yielding $P' = P \cup A$. For $p \in P$,*

$$\frac{\partial \ell'}{\partial s_p} - \frac{\partial \ell}{\partial s_p} = \frac{M}{K(K+M)}. \quad (11)$$

For $a \in A$, the difference is $-1/(K+M)$. For the remaining candidates, it is zero.

Proof. The scores, candidates, and weights are unchanged, so D and all q_j remain fixed. For an original positive, subtracting the two derivatives gives

$$(q_p - 1/(K+M)) - (q_p - 1/K) = \frac{M}{K(K+M)}. \quad (12)$$

A newly positive candidate changes target probability from zero to $1/(K+M)$. The remaining targets stay zero. The original positives jointly lose $M/(K+M)$ target probability, precisely the amount assigned to the new positives. $\square$

The premise that candidates are already present matters: adding new candidates changes the denominator as well. Likewise, changing weights changes q . Symmetric training can also change which text anchors have a positive image. None of these effects is included in the fixed-row identity.

H.3 A constant matching the normalizer

Proposition 2 (Matched negative mass). *Let N be nonempty under the common assumptions. Define*

$$c^* = \frac{\sum_{n \in N} w_n e^{s_n}}{\sum_{n \in N} e^{s_n}}. \quad (13)$$

Replace w_n by c^ for each $n \in N$, keeping positive weights one. At the fixed scores, the new normalizer, loss, each positive probability, and each positive derivative equal their original values. If c^* is held fixed during differentiation, the total negative derivative is also unchanged.*

Proof. The denominator defining c^* is positive because N is nonempty and every exponential is positive. Therefore c^* exists, is nonnegative, and satisfies

$$\sum_{n \in N} c^* e^{s_n} = \sum_{n \in N} w_n e^{s_n}. \quad (14)$$

The positive exponential mass is unchanged. Adding it to both sides proves equality of the normalizers. The target term also stays fixed, giving equal losses. Each positive probability is e^{s_p}/D , so its probability and derivative are equal. Finally, the total negative derivative is the negative exponential mass divided by D in each case. $\square$

Individual negative derivatives are generally unequal: they are $w_n e^{s_n}/D$ and $c^* e^{s_n}/D$. Matching preserves the total amount of competition, while changing its allocation. This is a row-specific analytical construction; a single global constant need not match every row. If the constant is recomputed as a function of scores without detachment, the derivative calculation is different.

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Source	17	0.0001	1	81.35	83.44	67.65	56.28	39.65
Source	29	0.0001	1	81.09	83.42	68.09	56.02	39.46
Source	43	0.0001	3	81.12	82.91	66.47	55.26	39.02
Expanded	17	0.0003	5	86.75	83.38	67.58	48.36	35.40
Expanded	29	0.0003	10	86.79	83.54	66.91	45.24	33.82
Expanded	43	0.0003	5	87.46	83.25	67.19	48.28	35.30
Contradiction	17	0.0003	5	87.04	83.33	67.58	48.46	35.48
Contradiction	29	0.0003	10	87.13	83.49	67.02	45.02	33.76
Contradiction	43	0.0003	5	87.66	83.17	67.31	48.28	35.18
Neutral	17	0.0003	5	86.42	83.60	68.26	48.10	35.54
Neutral	29	0.0003	5	86.40	83.49	67.35	47.84	35.50
Neutral	43	0.0003	5	87.17	83.09	67.37	48.10	35.12
Combined	17	0.0003	5	86.64	83.46	68.09	48.04	35.53
Combined	29	0.0003	5	86.65	83.37	67.19	47.82	35.51
Combined	43	0.0003	7	86.94	82.88	67.56	45.92	34.77
Random	17	0.001	3	86.04	80.96	65.48	44.84	29.33
Random	29	0.001	5	86.17	81.55	66.58	42.80	30.62
Random	43	0.001	3	86.58	82.36	65.73	42.64	29.90

Table 8: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

H.4 Suppression does not control every negative group

For one positive, let all negative weights lie in $[0, 1]$. Write

$$S = \sum_j e^{s_j}, \quad D = e^{s_+} + \sum_{n \in N} w_n e^{s_n}. \quad (15)$$

The inequality $D \leq S$ implies $q_+ = e^{s_+}/D \geq p_+ = e^{s_+}/S$. Hence total negative derivative satisfies

$$\sum_{n \in N} q_n = 1 - q_+ \leq 1 - p_+. \quad (16)$$

This bound does not hold for every subset of negatives. Consider exponential masses 1 for the positive, 1 for a chosen negative group, and 100 for other negatives. Weight the chosen group by 0.5 and the others by 0.001. The chosen group’s probability changes from

$$\frac{1}{102} \approx 0.009804 \quad \text{to} \quad \frac{0.5}{1 + 0.5 + 0.1} = 0.3125. \quad (17)$$

The total negative mass decreases, but that group’s mass increases. Contradiction weighting by two also lies outside the premise $w_n \leq 1$.

H.5 Summed-positive probability

Let $A = \sum_{p \in P} e^{s_p}$. A different objective is

$$\ell_{\text{set}} = \log D - \log A. \quad (18)$$

For $p \in P$, direct differentiation gives

$$\frac{\partial \ell_{\text{set}}}{\partial s_p} = q_p - \frac{e^{s_p}}{A} \leq 0, \quad (19)$$

because $D \geq A$. The attraction is allocated according to the positive scores, rather than equally across positives. This permits the highest-scoring positives to dominate the target distribution. The identity describes a distinct objective, not an equivalent implementation of Equation 1.

H.6 Parameter-dependent weights and score updates

For differentiable positive weights $w_j(\theta) > 0$, fixed targets, and parameter-dependent scores,

$$\begin{aligned} \nabla_{\theta} \ell = & \sum_j (q_j - y_j) \nabla_{\theta} s_j \\ & + \sum_j q_j \nabla_{\theta} \log w_j. \end{aligned} \quad (20)$$

The second term vanishes for detached weights. At exact zero weights, this logarithmic formula requires treating those entries as a fixed mask; it does not describe crossing a zero boundary. Even with detached weights, a parameter update combines the score Jacobians across every candidate and batch. A sign or magnitude bound for an individual score derivative is not a corresponding

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Smoothing	17	0.0001	1	82.04	83.61	68.59	55.26	38.99
Smoothing	29	0.0001	10	73.63	79.76	60.67	50.92	33.45
Smoothing	43	0.0001	1	82.03	83.48	68.64	55.36	38.81
Ranking	17	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	29	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	43	0.0001	0	79.47	81.21	68.95	56.32	39.35
Fixed	17	0.0003	6	79.74	82.97	66.09	50.34	31.88
Fixed	29	0.0003	0	79.47	81.21	68.95	56.32	39.35
Fixed	43	0.0003	0	79.47	81.21	68.95	56.32	39.35
Shuffled	17	0.0003	1	82.85	83.10	66.64	52.60	38.10
Shuffled	29	0.0003	1	83.97	83.81	66.89	52.14	38.14
Shuffled	43	0.0003	1	83.20	83.54	67.16	52.06	38.33
Median	17	0.0001	3	82.76	83.54	67.96	53.64	38.90
Median	29	0.0001	2	82.44	83.74	67.96	54.42	39.42
Median	43	0.0001	2	82.78	83.89	68.09	53.86	39.30
Constant	17	0.0001	1	82.00	83.88	68.59	55.36	39.48
Constant	29	0.0001	1	82.03	83.65	68.70	55.10	39.39
Constant	43	0.0001	2	82.43	83.48	67.82	55.16	39.31

Table 9: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

Comparator	Endpoint	Δ	99.17% interval	Seed 17	Seed 29	Seed 43
Expanded	e-SNLI-VE	-0.255	[-0.605, +0.065]	-0.109	-0.137	-0.520
Expanded	SugarCrepe	-0.155	[-0.509, +0.209]	+0.080	-0.173	-0.373
Neutral	e-SNLI-VE	+0.081	[-0.100, +0.273]	+0.221	+0.253	-0.230
Neutral	SugarCrepe	-0.155	[-0.371, +0.056]	-0.133	-0.120	-0.213
Contradiction	e-SNLI-VE	-0.535	[-0.890, -0.193]	-0.398	-0.484	-0.723
Contradiction	SugarCrepe	-0.093	[-0.442, +0.261]	+0.133	-0.120	-0.293

Table 10: Six primary contrasts: Combined minus the comparator, in percentage points. Intervals use paired image-cluster resampling and the fixed six-comparison Bonferroni family.

bound on the parameter-gradient norm or on the score after optimization.

H.7 Other fixed-score identities

The following facts collect the single-positive special cases behind similarity suppression. Let $0 \leq w_n \leq 1$ for negatives, $w_+ = 1$, and

$$R = \sum_{n \in N} (1 - w_n) e^{s_n}, \quad D = S - R. \quad (21)$$

Since $0 \leq R < S$,

$$\log q_+ - \log p_+ = -\log(1 - R/S) \geq R/S \geq 0. \quad (22)$$

The logarithm inequality follows by integrating $1/(1-x) \geq 1$ over $[0, R/S]$. It compares probabilities at fixed scores and does not change their raw

ranking. For a group of m negatives at common score a and common weight w , its gradient mass is

$$G = \frac{mwe^a}{D}. \quad (23)$$

The denominator also depends on the group. Expressions with a denominator of the form $A + me^a$ saturate as m increases; they do not imply an unbounded linear gain from multiplicity. At equal scores, a smaller weight gives smaller normalized probability within the same denominator. This says nothing about whether the weighting identifies a correct semantic relation.

As every weight tends to one, the weighted loss converges to the ordinary loss by continuity. A common negative weight below one does not yield the ordinary loss while the positive weight remains

Policy	Add A	Add O	Repl. A	Repl. O	Repl. R	Swap A	Swap O
Frozen	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Source	83.24	89.72	83.76	94.09	72.00	67.47	61.77
Expanded	87.86	90.74	82.91	93.02	72.00	65.12	60.68
Contradiction	88.05	90.59	83.04	92.88	72.02	64.96	60.27
Neutral	87.91	90.98	82.91	93.18	71.46	65.17	60.41
Combined	87.91	90.74	83.04	93.20	71.08	65.27	59.05
Random	86.85	89.19	81.09	92.39	68.66	63.06	57.14
Smoothing	80.54	87.20	83.46	93.48	72.43	67.62	62.99
Ranking	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Fixed	80.11	87.86	82.66	93.79	69.77	66.97	61.36
Shuffled	86.37	90.88	83.08	93.68	70.63	68.07	61.36
Median	85.12	91.21	83.71	94.05	71.03	68.22	62.31
Constant	84.20	90.80	83.46	94.23	72.00	67.57	62.31

Table 11: SugarCrepE category means (percent). A, O, and R denote attribute, object, and relation. Category denominators are listed in Appendix B.

Seed	Queries	Pool	Lost	New	Net loss
17	I→T	5,000	649	253	396
17	T→I	25,000	2,195	1,132	1,063
29	I→T	5,000	832	293	539
29	T→I	25,000	2,651	1,241	1,410
43	I→T	5,000	597	248	349
43	T→I	25,000	1,976	1,045	931

Table 12: Retrieval success transitions from source positives to positive expansion. Lost and new count changed rank-one successes on the identical query pool.

one. Finally, weighted normalization is ordinary softmax normalization on shifted scores $s_j + \log w_j$, with $\log 0 = -\infty$. It is not label smoothing, which changes the targets.

Policy	I→T R1	R5	R10	T→I R1	R5	R10
Frozen	56.32	79.82	87.12	39.35	65.40	75.62
Source	55.85	79.59	86.93	39.38	65.67	75.91
Expanded	47.29	71.56	80.89	34.84	60.77	71.50
Contradiction	47.25	71.61	80.87	34.81	60.78	71.46
Neutral	48.01	72.58	81.79	35.39	61.56	72.19
Combined	47.26	71.93	81.33	35.27	61.36	72.05
Random	43.43	67.84	77.81	29.95	55.28	66.61
Smoothing	53.85	77.94	85.71	37.08	63.47	73.72
Ranking	56.32	79.82	87.12	39.35	65.40	75.62
Fixed	54.33	77.83	85.75	36.86	62.66	73.19
Shuffled	52.27	76.63	84.83	38.19	64.41	74.58
Median	53.97	77.93	85.83	39.21	65.42	75.66
Constant	55.21	78.85	86.55	39.39	65.62	75.85

Table 13: Full-pool COCO recall at ranks one, five, and ten, averaged across selected seeds (percent).

Input	Probabilities	Acc.	Macro-F1	ECE	Brier	NLL
Image	uncalibrated	35.64	34.06	4.73	0.674	1.111
Image	calibrated	35.64	34.06	0.05	0.666	1.098
Text	uncalibrated	61.76	61.38	5.78	0.509	0.879
Text	calibrated	61.76	61.38	0.68	0.503	0.858
Multimodal	uncalibrated	65.69	64.96	12.53	0.487	0.887
Multimodal	calibrated	65.69	64.96	1.38	0.457	0.783

Table 14: Held-out relation classification on the newly sampled test set. Accuracy, macro-F1, and ECE are percentages; Brier and NLL retain their native scale.

Input	Gate	Accept	Correct	Coverage	Precision	95% interval
Image	supported	0	0	0.00	n/a	n/a
Image	contradicted	0	0	0.00	n/a	n/a
Text	supported	0	0	0.00	n/a	n/a
Text	contradicted	0	0	0.00	n/a	n/a
Multimodal	supported	0	0	0.00	n/a	n/a
Multimodal	contradicted	683	647	11.65	94.73	[92.72, 96.59]

Table 15: Selective test predictions. Precision is undefined when no pair is accepted. Intervals resample whole test images and are descriptive.

Input	C	Temp.	Max. iter.	OOF F1
Image	0.1	3.085	1182	33.17
Text	1	1.312	2318	63.46
Multimodal	0.01	1.904	2505	67.22

Table 16: Classifier selection and completion. Maximum iteration count is over the full-training grid; OOF is image-disjoint out-of-fold macro-F1 (percent).

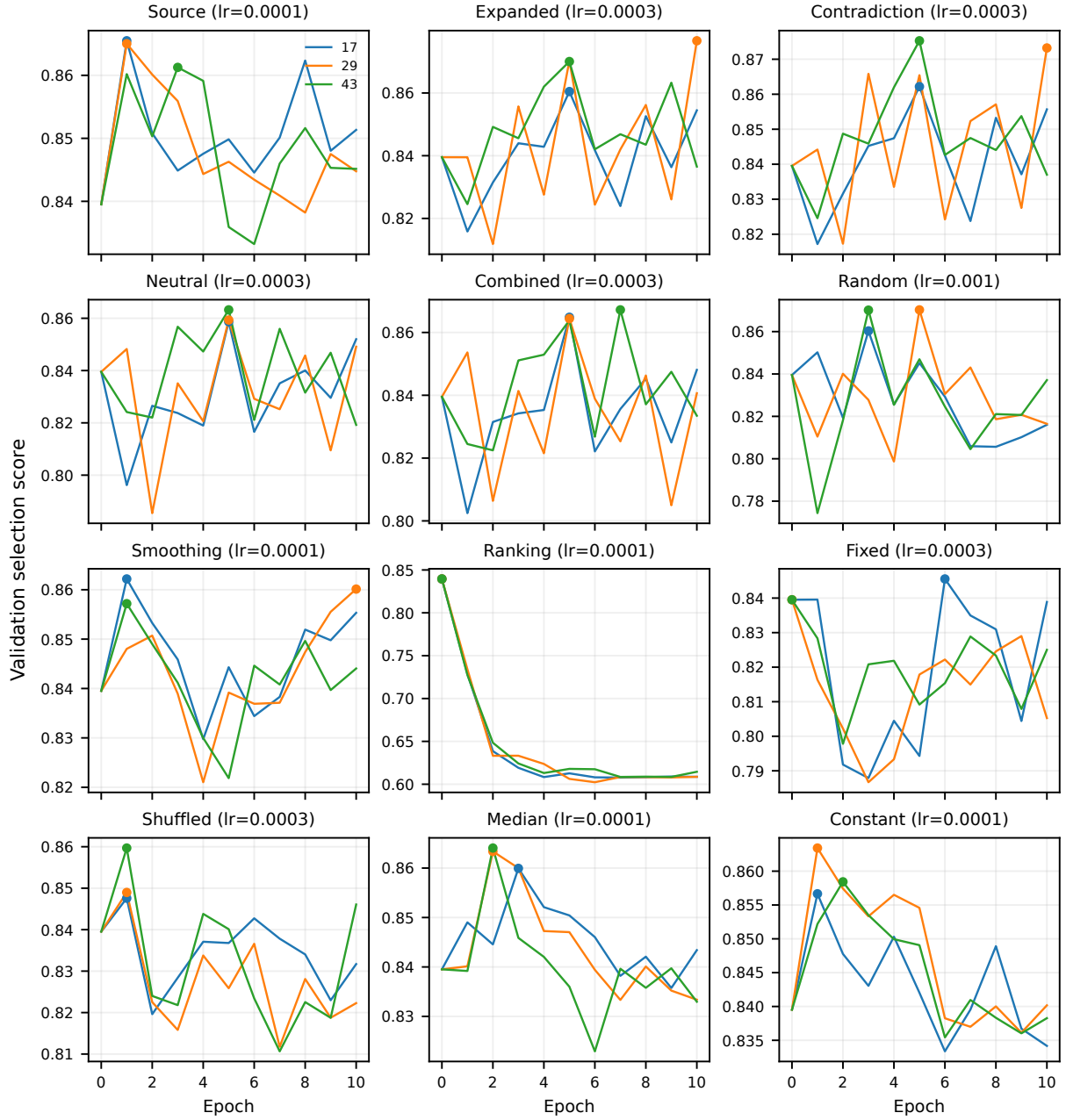


Figure 3: Validation trajectories at the selected learning rate of each policy. Lines are seeds; dots mark retained epochs. Every candidate evaluated epoch zero through ten. These are selection data, not held-out outcomes.